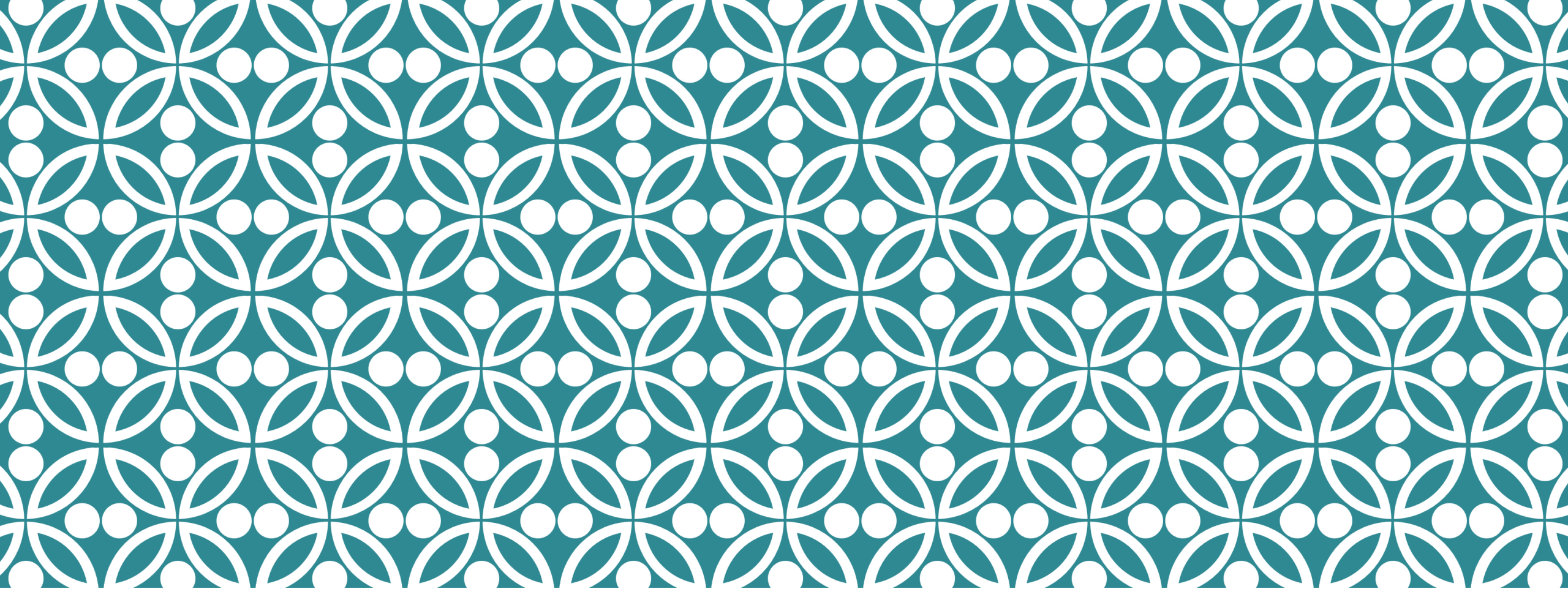




SESSION 9: MACHINE LEARNING FOR TEXT

DR. SOFIA GIL-CLAVEL

- ❖ Introduction to lists
- ❖ Recap on Text Analysis and Unsupervised Methods
- ❖ Unsupervised Methods for Text
- ❖ Recap on Statistics and Supervised Methods
- ❖ Supervised Methods for Text

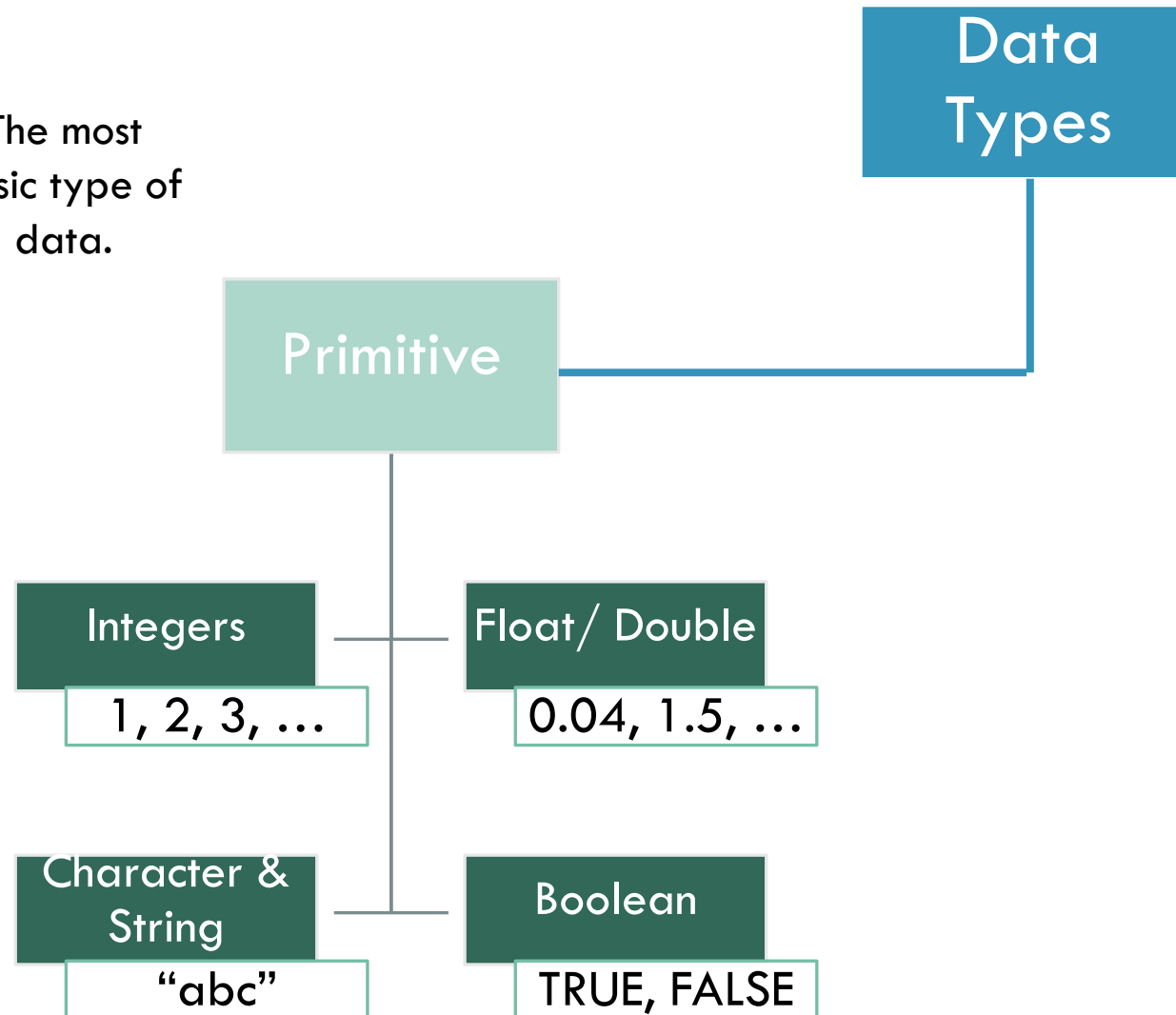
1. INTRODUCTION TO LISTS

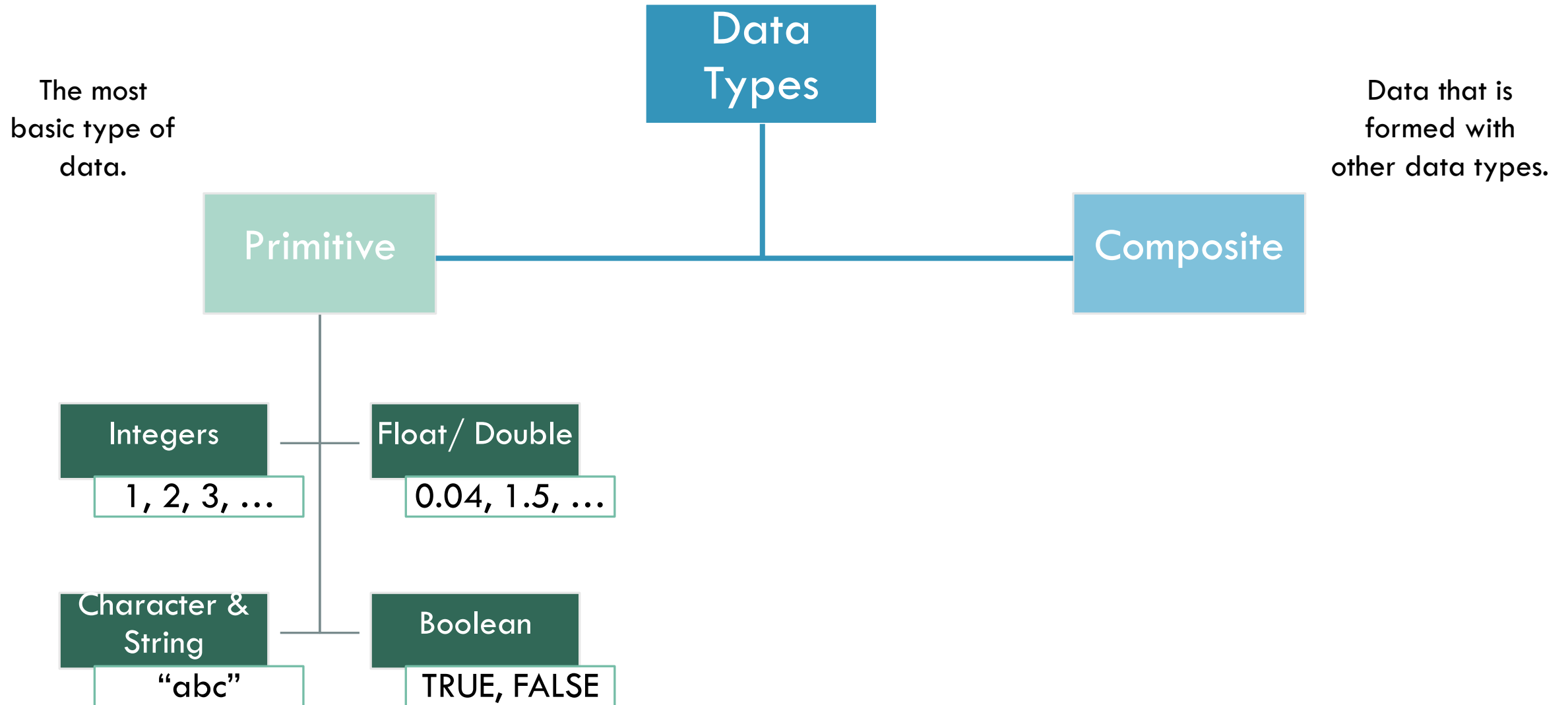
Data Types

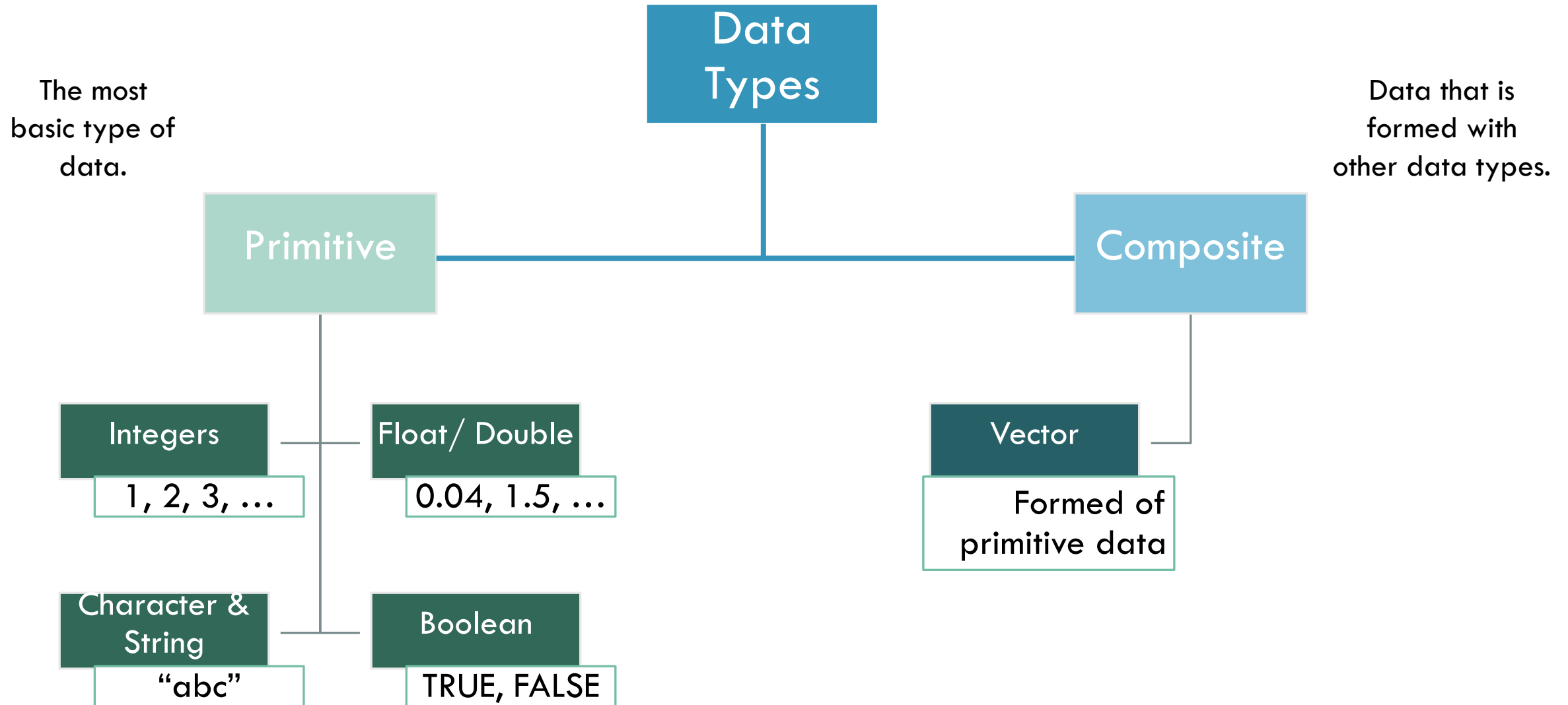
The most
basic type of
data.

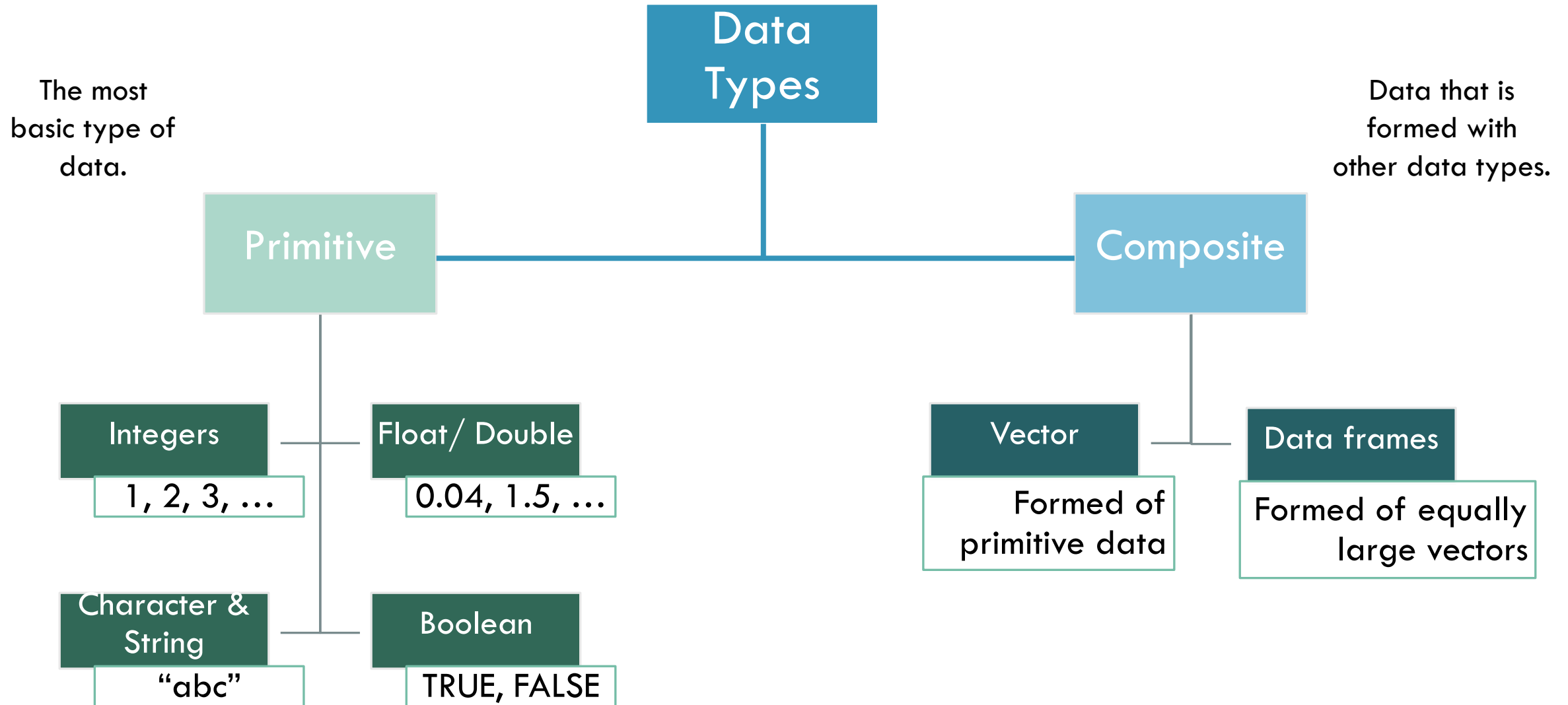


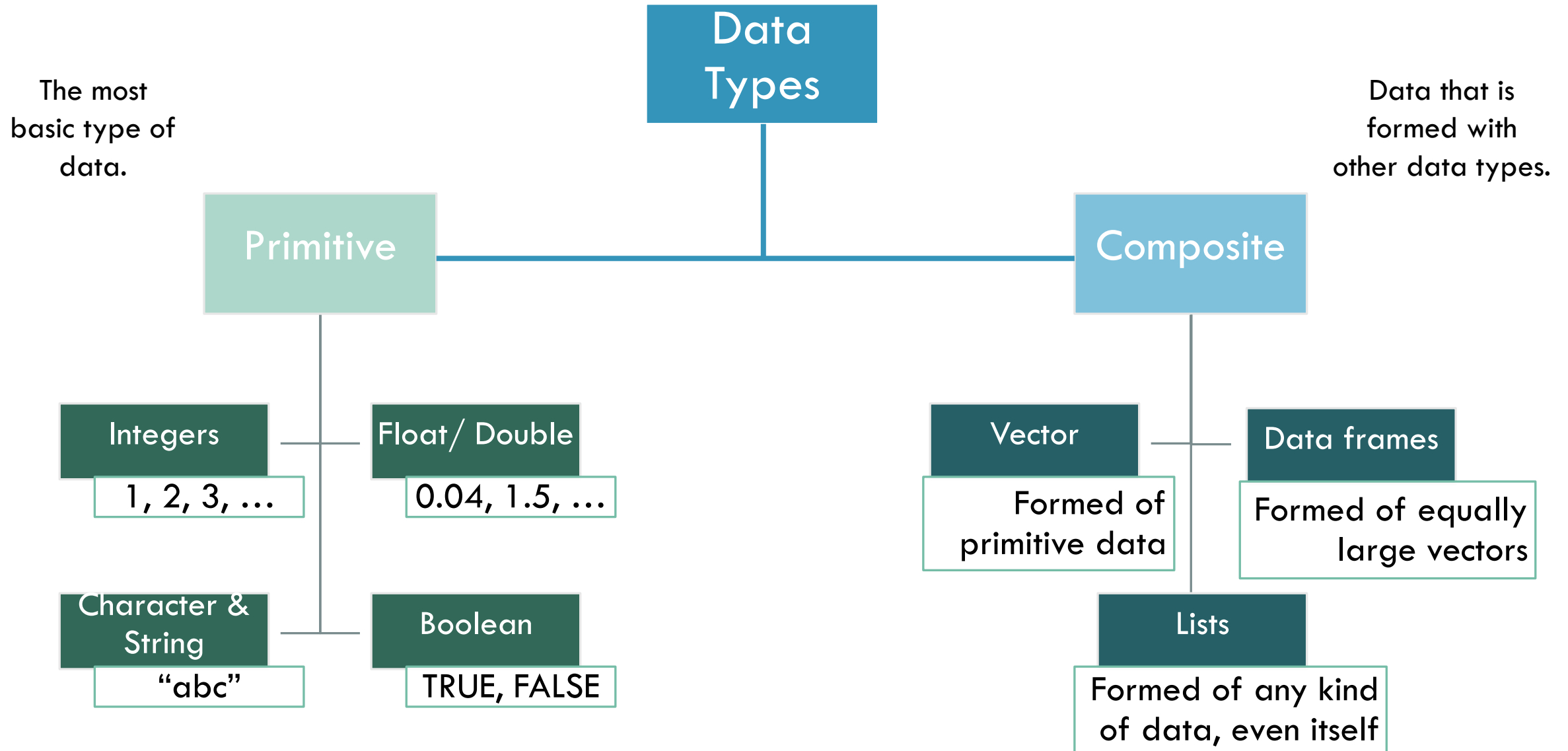
The most
basic type of
data.











LISTS

A list is a generic vector that contains different types of objects of different sizes.

```
n = c(2, 3, 5)
s = c("aa", "bb", "cc", "dd", "ee")
b = c(TRUE, FALSE, TRUE, FALSE, FALSE)
x = list(n, s, b, 3)  # x contains copies of n, s, b
```

Generic way to
generate a list.

```
> v = list(bob=c(2, 3, 5), john=c("aa", "bb"))
> v
$bob
[1] 2 3 5

$joh
[1] "aa" "bb"
```

Assigning names
to members of a
list.

ELEMENT EXTRACTION AND ALTERATION

Extracting elements from the list:

```
> x[2]
[[1]]
[1] "aa" "bb" "cc" "dd" "ee"

> x[[2]][1] = "ta"
> x[[2]]
[1] "ta" "bb" "cc" "dd" "ee"
```

Extracting elements from a list with names:

```
> v["bob"]
$bob
[1] 2 3 5
```

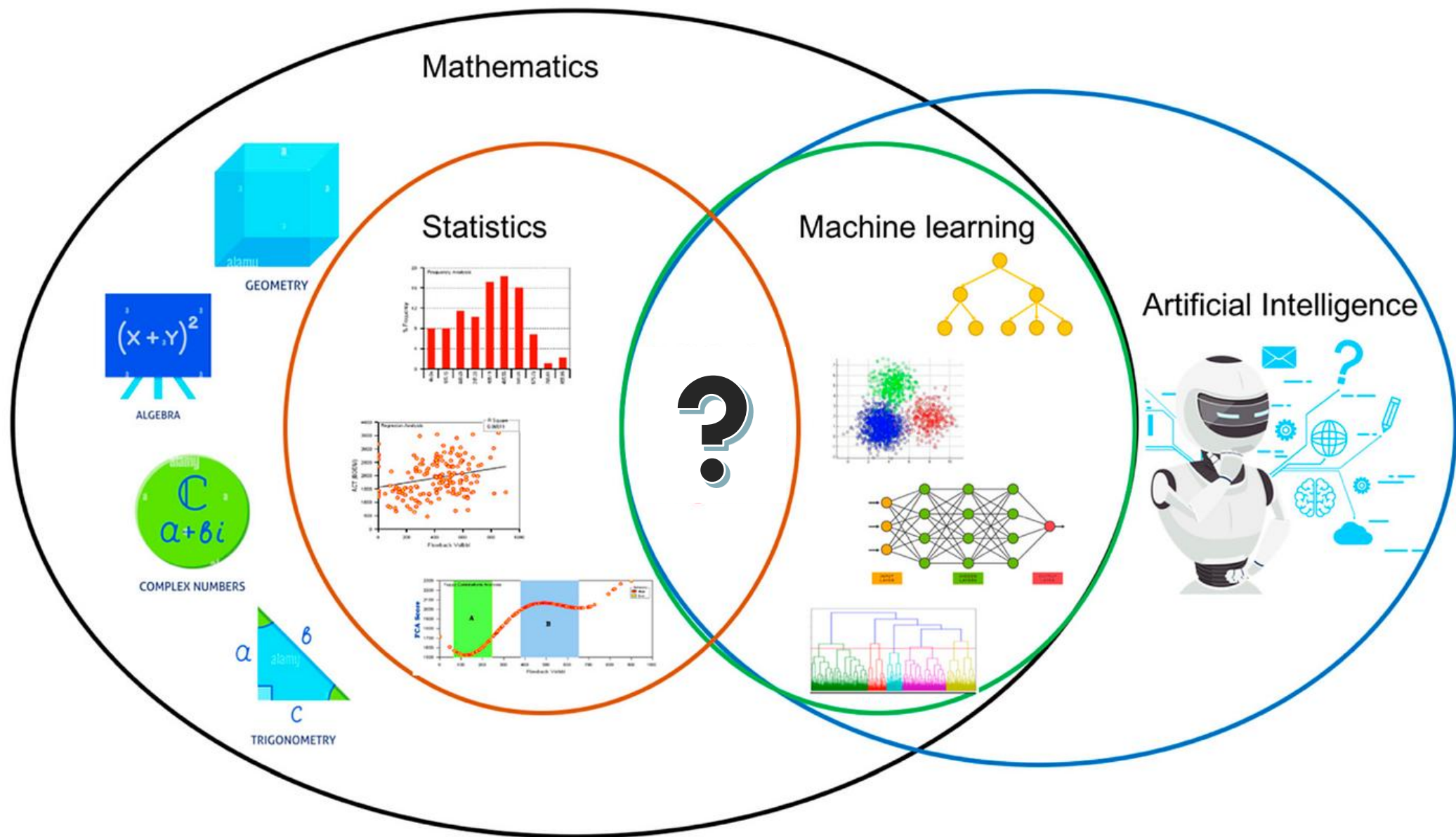
```
> v[["bob"]]
[1] 2 3 5
```

```
> v$bob
[1] 2 3 5
> attach(v)
> bob
[1] 2 3 5
> detach(v)
```

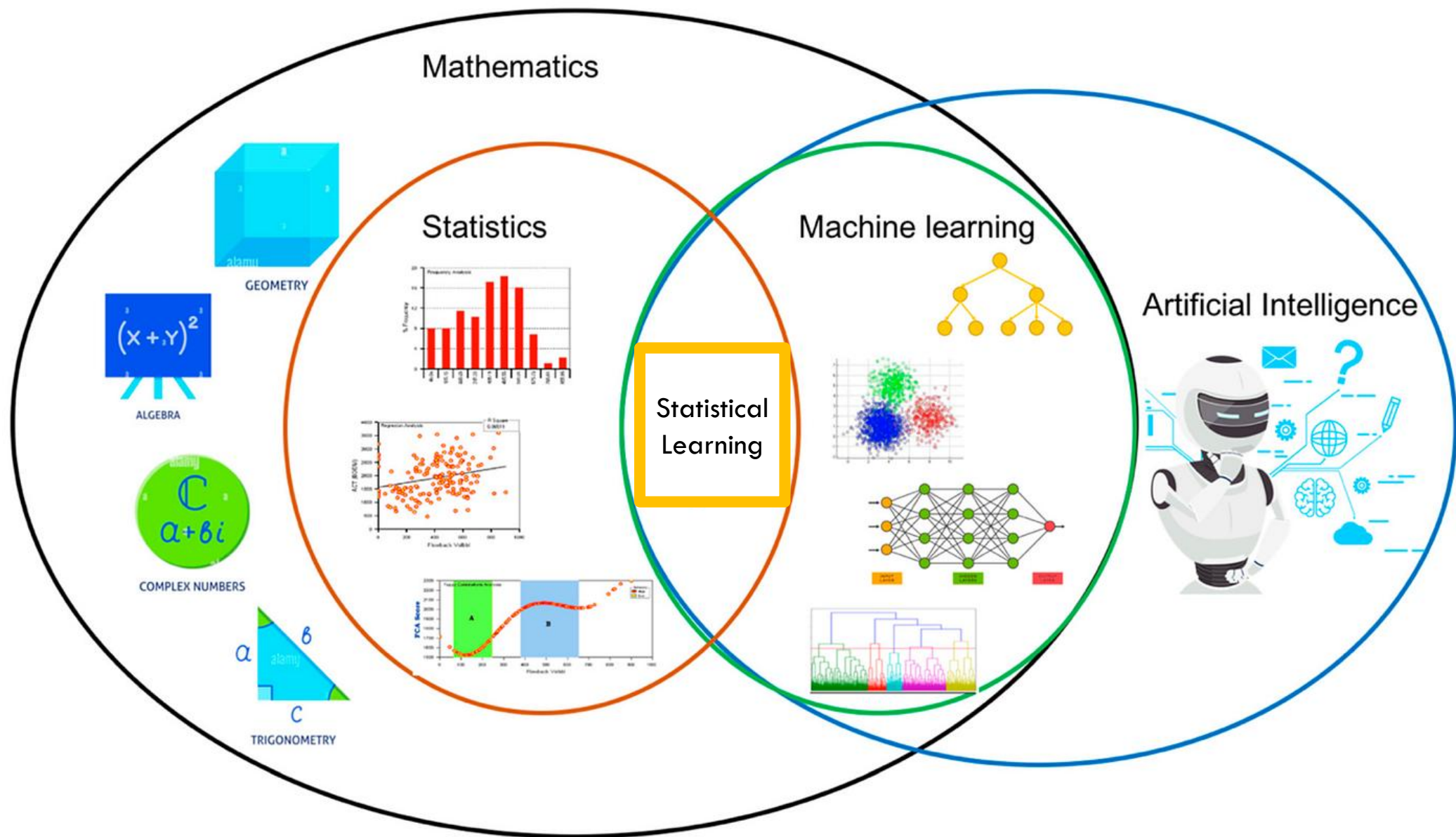
2. PRELIMINARIES FOR MACHINE LEARNING ON TEXT

1. Recap on Statistical Learning
2. Unsupervised vs. Supervised
3. From text to numbers

2.1 STATISTICAL LEARNING



Original source: <https://medium.com/stats-learning/differences-and-synergies-between-statistics-and-machine-learning-90cea85d4cf5>



Original source: <https://medium.com/stats-learning/differences-and-synergies-between-statistics-and-machine-learning-90cea85d4cf5>

WHY STATISTICAL LEARNING?

So far in these workshops we have learned to handle data frames in R.

data frame
name

columns

rows

variable

value/element

```
> Bikeshare
```

	bikers	season	day	holiday	weekday	workingday	temp	atemp	hum
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	16	1	1	0	6	0	0.24	0.288	0.81
2	40	1	1	0	6	0	0.22	0.273	0.8
3	32	1	1	0	6	0	0.22	0.273	0.8
4	13	1	1	0	6	0	0.24	0.288	0.75
5	1	1	1	0	6	0	0.24	0.288	0.75
6	1	1	1	0	6	0	0.24	0.258	0.75
7	2	1	1	0	6	0	0.22	0.273	0.8
8	3	1	1	0	6	0	0.2	0.258	0.86
9	8	1	1	0	6	0	0.24	0.288	0.75
10	14	1	1	0	6	0	0.32	0.348	0.76

10,055 more rows
3 more variables: windspeed <dbl>, casual <dbl>,
registered <dbl>
Use `print(n = ...)` to see more rows

WHY STATISTICAL LEARNING?

So far in these workshops we have learned to handle data frames in R. This is very convenient, as we can analyze these data frames using statistical symbolic language!

- **Dependent (Y)**: Also known as response variable.
- **Independent (X)**: Also known as input, predictor, feature, or just variable.

```
> Bikeshare
# A tibble: 8,645 × 12
```

	bikers	season	day	holiday	weekday	workingday	temp	atemp	hum
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	16	1	1	0	6	0	0.24	0.288	0.81
2	40	1	1	0	6	0	0.22	0.273	0.8
3	32	1	1	0	6	0	0.22	0.273	0.8
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9	8	1	1	0	6	0	0.24	0.288	0.75
10	14	1	1	0	6	0	0.32	0.348	0.76

```
# 8,645 more rows
# 3 more variables: windspeed <dbl>, casual <dbl>
#   registered <dbl>
# Use `print(n = ...)` to see more rows
```

dependent variable (Y)

independent variables (X)

WHY STATISTICAL LEARNING?

So far in these workshops we have learned to handle data frames in R. This is very convenient, as we can analyze these data frames using statistical symbolic language!

Symbolic
Language

➤ **Dependent** (Y): Also known as response variable.

➤ **Independent** (X): Also known as input, predictor, feature, or just variable.

$$Y \sim X1 + X2 + X3$$

 was built to
understand symbolic
language!

2.2 UNSUPERVISED VS. SUPERVISED

SUPERVISED STATISTICAL LEARNING

In supervised learning, for each observation of the predictor measurement(s) X_i there is an associated response measurement Y_i .

$$Y_i \sim X1_i + X2_i + X3_i$$

i refers to the data frame row

We wish to fit a model that relates the response to the predictors, with the aim of:

- Prediction: accurately predicting the response for future observations.
- Inference: better understanding the relationship between the response and the predictors.

Many classical statistical learning methods such as linear regression and logistic regression, as well as more modern approaches such as GAM, boosting, and support vector machines, operate in the supervised learning domain.

UNSUPERVISED STATISTICAL LEARNING

Unsupervised learning describes the somewhat more challenging situation in which for every observation $i = 1, \dots, n$, we observe a vector of measurements X_i but no associated response Y_i .

$$\text{No } Y_i \sim X1_i + X2_i + X3_i$$

We are not interested in prediction, because we do not have an associated response variable Y . The goal is to discover interesting things about the measurements on $X1, X2, \dots, Xp$. Is there an informative way to visualize the data? Can we discover subgroups among the variables or among the observations?

Unsupervised learning is often performed as part of an exploratory data analysis.

2.3 SETTING UP A TEXT RECIPE

SETTING UP A TEXT RECIPE

textrecipes 1.1.0



textrecipes

Introduction

textrecipes contain extra steps for the [recipes](#) package for preprocessing text data.

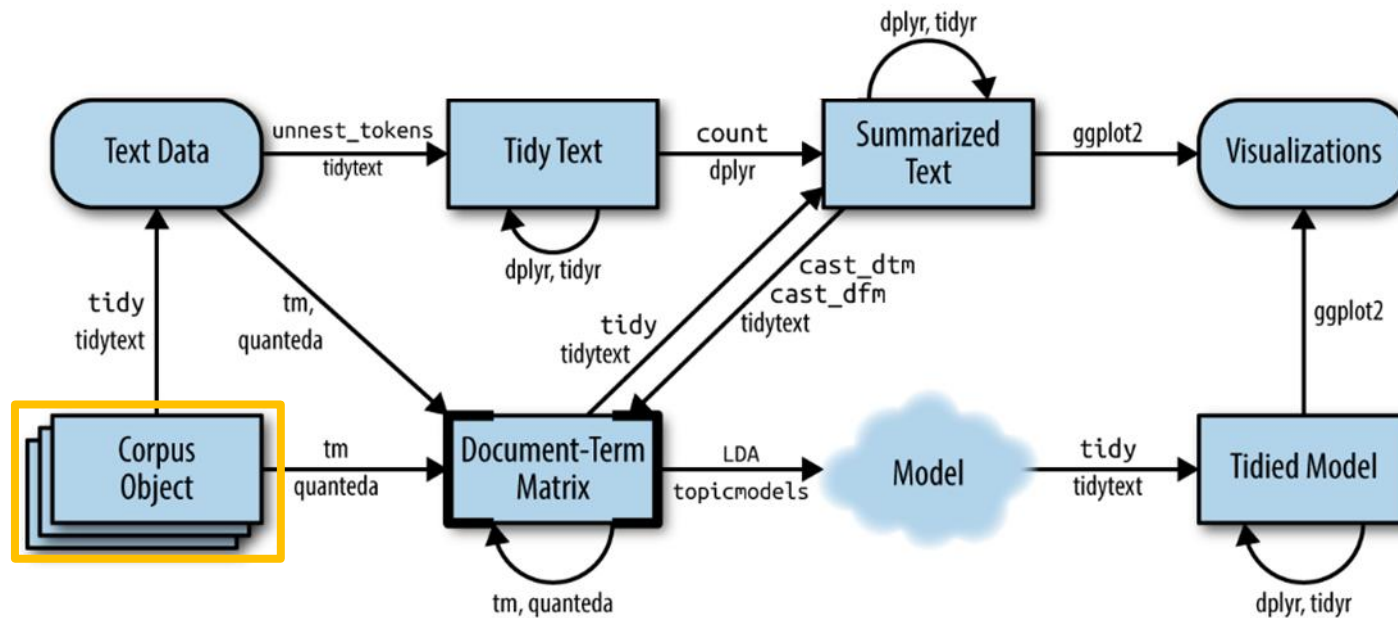
A recipe contains:

1. The text processing steps:
 - Tokenizing
 - Stopword removal
 - Stemming or lemmatizing (requires SpaCy)
 - TF-IDF transformation
- A. The unsupervised learning method.
- B. The model to train the supervised learning method.

3. UNSUPERVISED METHODS FOR TEXT

- 1. Topic Modelling
- 3. Latent Dirichlet allocation (LDA)

THE PIPELINE



THE DATA

We will use the data “acq” from the package “tm”:

This dataset holds **50 news articles** with additional meta information from the Reuters-21578 data set. **All documents belong to the topic acq dealing with corporate acquisitions.**

```
library(tm)  
data("acq")
```

THE DATA

A corpus is a collection of unstructured documents or texts.

In the case of our data, the corpus corresponds to 50 articles about corporate acquisition.

Name	Type	Value
acq	list [50] (S3: VCorpus, Corpus)	List of length 50
10	list [2] (S3: PlainTextDocument, T	List of length 2
content	character [1]	'Computer Terminal Systems Inc said\nit
meta	list [15] (S3: TextDocumentMeta)	List of length 15
12	list [2] (S3: PlainTextDocument, T	List of length 2
content	character [1]	'Ohio Mattress Co said its first\nquarter,
meta	list [15] (S3: TextDocumentMeta)	List of length 15

To turn the list
into a data
frame.



```
ACQ<-data.frame(text=unlist(sapply(acq, `[`, "content")),  
                  stringsAsFactors=FALSE)
```



1. TEXT PROCESSING STEPS ~~1~~ ~ $X1_i + X2_i + X3_i$

`ACQ_rec <- recipe(~ text, data = ACQ) %>%`

`# 1. Break text into individual words (tokens)`

`step_tokenize(text) %>%`

`# 2. REMOVE STOP WORDS FIRST (while tokens are still single words)`

`step_stopwords(text, keep = FALSE)%>%`

`# 3. BUILD N-GRAMS SECOND (from the remaining meaningful words)`

`step_ngram(text, num_tokens = 2, min_num_tokens = 1) %>%`

`# 4. Filter or convert to features`

`step_tokenfilter(text, max_tokens = 1000)`

3.1 TOPIC MODELING

WHAT IS TOPIC MODELING?

Topic modeling is a collection of text-mining techniques that uses statistical and machine learning models to **automatically discover hidden abstract topics** in a collection of documents.

Topic modeling is also an amalgamation of a set of **unsupervised techniques** that's capable of detecting word and phrase patterns within documents and **automatically cluster word groups** and similar expressions helping in best representing a set of documents.

Source: <https://www.scaler.com/topics/nlp/topic-modelling-in-natural-language-processing/>

ARE PCA, TOPIC MODELING, AND CLUSTERING THE SAME?

All three algorithms, PCA, topic modeling, and clustering are unsupervised and used to **simplify the data set with a small number of summaries**, the major difference lies in how they are used:

- PCA looks to find a low-dimensional representation of the observations that explain a good fraction of the variance. Since output representations are unrelated, they can also be used as **input to clustering**.
- Topic modeling and PCA both can be used for dimensionality reduction, but LDA provides **better accuracies and explainability in terms of text**.
- Clustering looks to find homogeneous subgroups among the observations by **maximizing the distance** between clusters, clusters are not known in advance.

Source: <https://www.scaler.com/topics/nlp/topic-modelling-in-natural-language-processing/>

METHODS FOR DIMENSIONALITY REDUCTION AND CLUSTERING

Here we can also perform Dimensionality Reduction! But in this case, there are other well-known methods:

- Non-Negative Matrix Factorization
- LDA – Latent Dirichlet Allocation
- LSA – Latent Semantic Allocation
- PLSA – Probabilistic Latent Semantic Analysis
- Ida2vec – Deep Learning Model
- tBERT – Topic BERT

Source: <https://www.scaler.com/topics/nlp/topic-modelling-in-natural-language-processing/>

METHODS FOR DIMENSIONALITY REDUCTION AND CLUSTERING

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- PLSA – Probabilistic Latent Semantic Analysis
- Ida2vec – Deep Learning Model
- tBERT – Topic BERT



We will focus on this one!

Source: <https://www.scaler.com/topics/nlp/topic-modelling-in-natural-language-processing/>

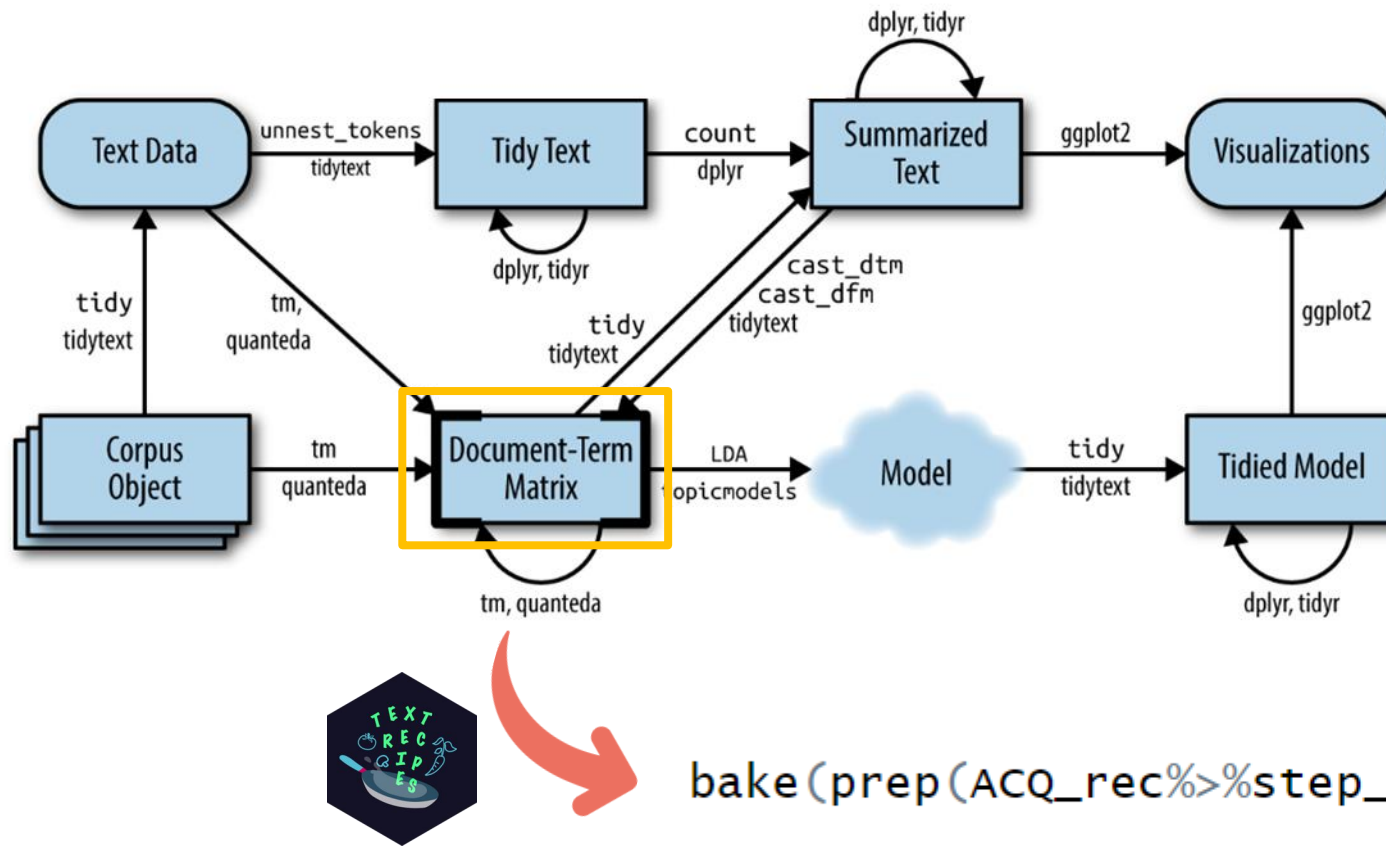
2.2 LATENT DIRICHLET ALLOCATION

LATENT DIRICHLET ALLOCATION

Latent Dirichlet allocation is one of the most common algorithms for topic modeling. Without diving into the math behind the model, we can understand it as being guided by two principles.

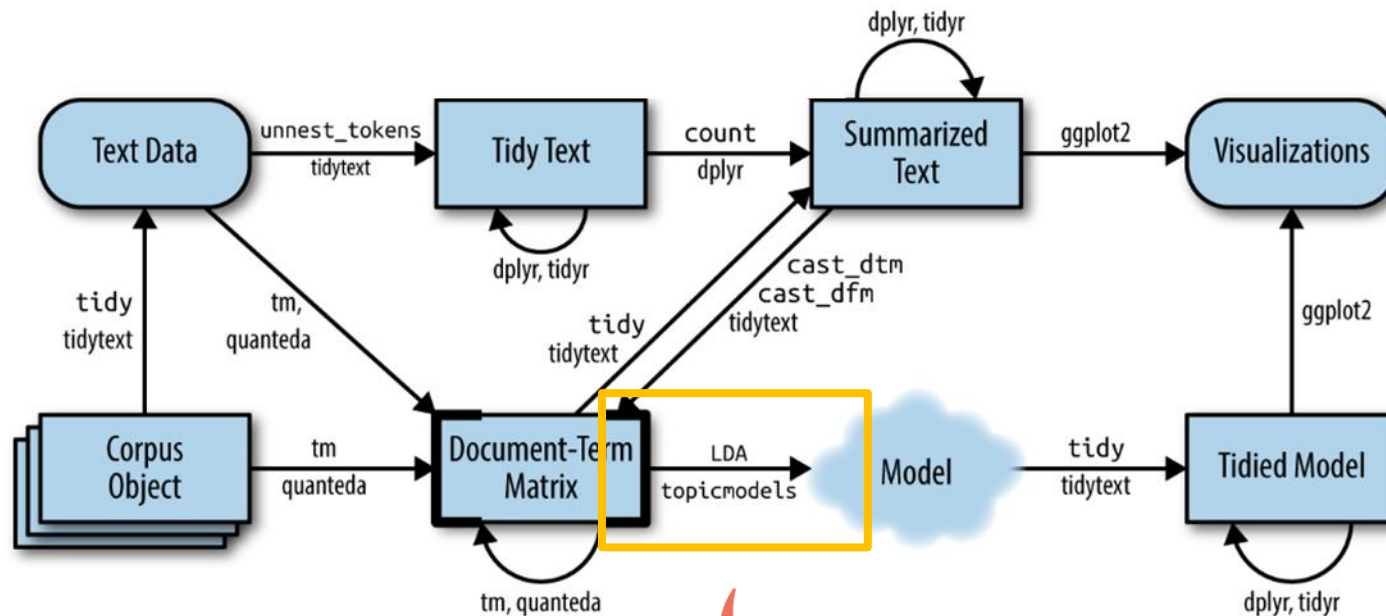
- **Every document is a mixture of topics.** We imagine that each document may contain words from several topics in particular proportions. For example, in a two-topic model we could say “Document 1 is 90% topic A and 10% topic B, while Document 2 is 30% topic A and 70% topic B.”
- **Every topic is a mixture of words.** For example, we could imagine a two-topic model of American news, with one topic for “politics” and one for “entertainment.” The most common words in the politics topic might be “President”, “Congress”, and “government”, while the entertainment topic may be made up of words such as “movies”, “television”, and “actor”. Importantly, words can be shared between topics; a word like “budget” might appear in both equally.

LATENT DIRICHLET ALLOCATION



 `bake(prepare(ACQ_rec%>%step_tf(text)), new_data = NULL)`

LATENT DIRICHLET ALLOCATION



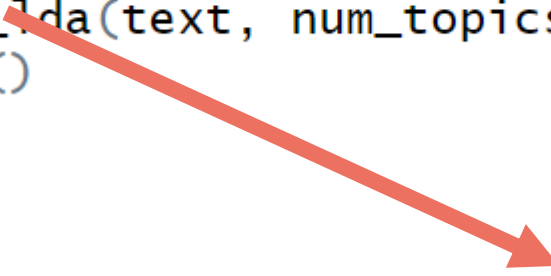
```
ACQ_obj <- ACQ_rec %>%  
  # 5. lda  
  step_lda(text, num_topics = 4) %>%  
  prep()
```

EXTRACTING RESULTS

We can extract the results from each recipe step using the step number to extract that element from the list.



```
ACQ_obj<-ACQ_rec%>%  
# 5.1lda  
step_lda(text, num_topics = 4)%>%  
prep()
```



```
ACQ_LDA<-ACQ_obj$steps[[5]]
```

Let's check the top ten words per cluster!



```
ACQ_LDA$get_top_words(n=10)
```

```
ACQ_LDA$plot()
```

OTHER IMPORTANT VALUES

Some important attributes acquired from the LDA Model:

- **phi** : Posterior per-topic-per-word probabilities
- **theta** : Posterior per-document-per-topic probabilities
- **alpha** : Prior per-document-per-topic probabilities
- **beta** : Prior per-document-per-topic probabilities
- **coherence** : The probabilistic coherence of each topic

EXTRACTING PHI

```
# Pull raw matrix and vocabulary names from the private environment
phi_matrix_raw <- ACQ_LDA$.__enclos_env__$private$components_
vocab_names    <- ACQ_LDA$.__enclos_env__$private$vocabulary

# Format into a tidy data frame of probabilities
rownames(phi_matrix_raw) <- paste0("Topic ", 1:nrow(phi_matrix_raw))
colnames(phi_matrix_raw) <- vocab_names

phi_matrix_probs <- t(phi_matrix_raw / rowSums(phi_matrix_raw))
```


LDA doesn't specifically inform us about what each topic is about. By looking at the representative words of each topic, we as the expert will give meaning to each topic.

Topic 1



Topic 2



Topic 3



Topic 4



3. PRELIMINARIES FOR SUPERVISED LEARNING

- 1. Recap on Statistics
- 2. Regression vs. Classification
- 3. Prediction vs. Classification

3.1 RECAP ON STATISTICS

SOME NOTATION

```
> Bikeshare
# A tibble: 8,645 x 12
```

	bikers	season	day	holiday	weekday	workingday	temp	atemp	hum
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	16	1	1	0	6	0	0.24	0.288	0.81
2	40	1	1	0	6	0	0.22	0.273	0.8
3	32	1	1	0	6	0	0.22	0.273	0.8
4	13	1	1	0	6	0	0.24	0.288	0.75
5	1	1	1	0	6	0	0.24	0.288	0.75
6	1	1	1	0	6	0	0.24	0.258	0.75
7	2	1	1	0	6	0	0.22	0.273	0.8
8	3	1	1	0	6	0	0.2	0.258	0.86
9	8	1	1	0	6	0	0.24	0.288	0.75
10	14	1	1	0	6	0	0.32	0.348	0.76

```
# 8,645 more rows
# 3 more variables: windspeed <dbl>, casual <dbl>
#   registered <dbl>
# Use `print(n = ...)` to see more rows
```

$$Y_i \sim X1_i + X2_i + X3_i \sim f(X_i)$$

dependent
variable (Y)

independent
variables (X)

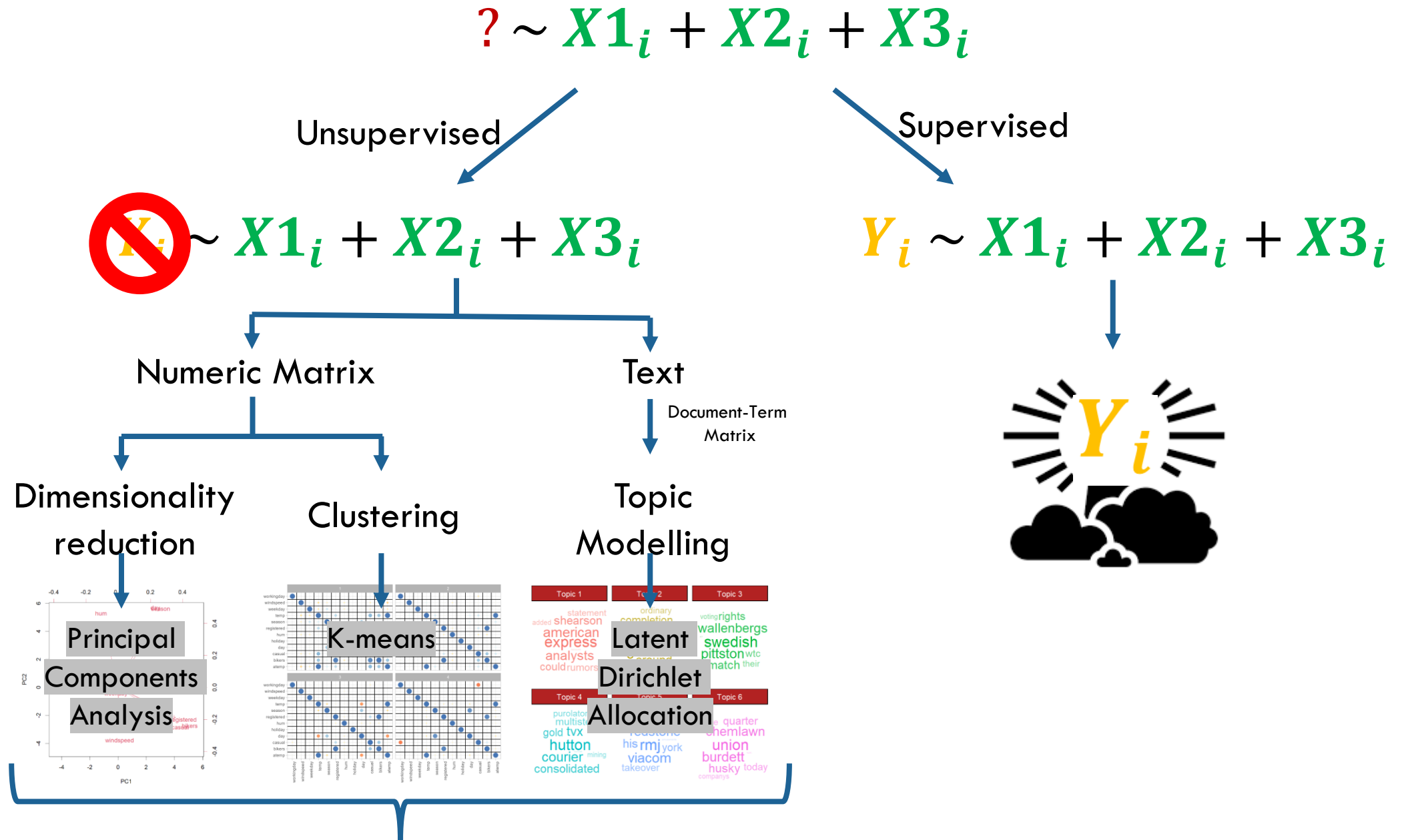
This value comes
from our data.

We estimate the model (f) parameters
 $\hat{\beta}_i$ and use them to predict $\hat{Y}_i$

$$\hat{\beta}_1 X1_i + \hat{\beta}_2 X2_i + \hat{\beta}_3 X3_i = \hat{Y}_i$$

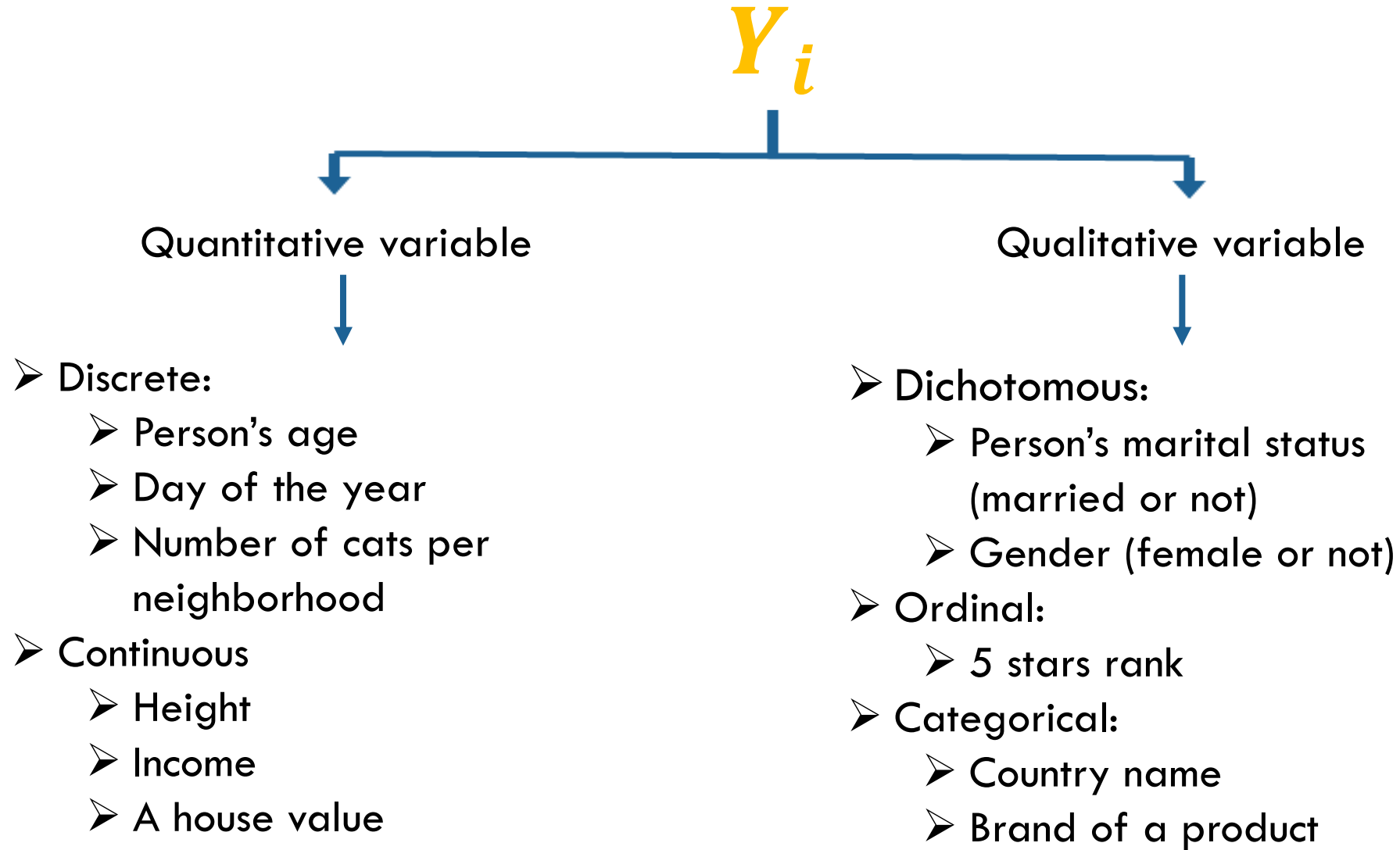
This value is the
prediction.

UNSUPERVISED VS. SUPERVISED

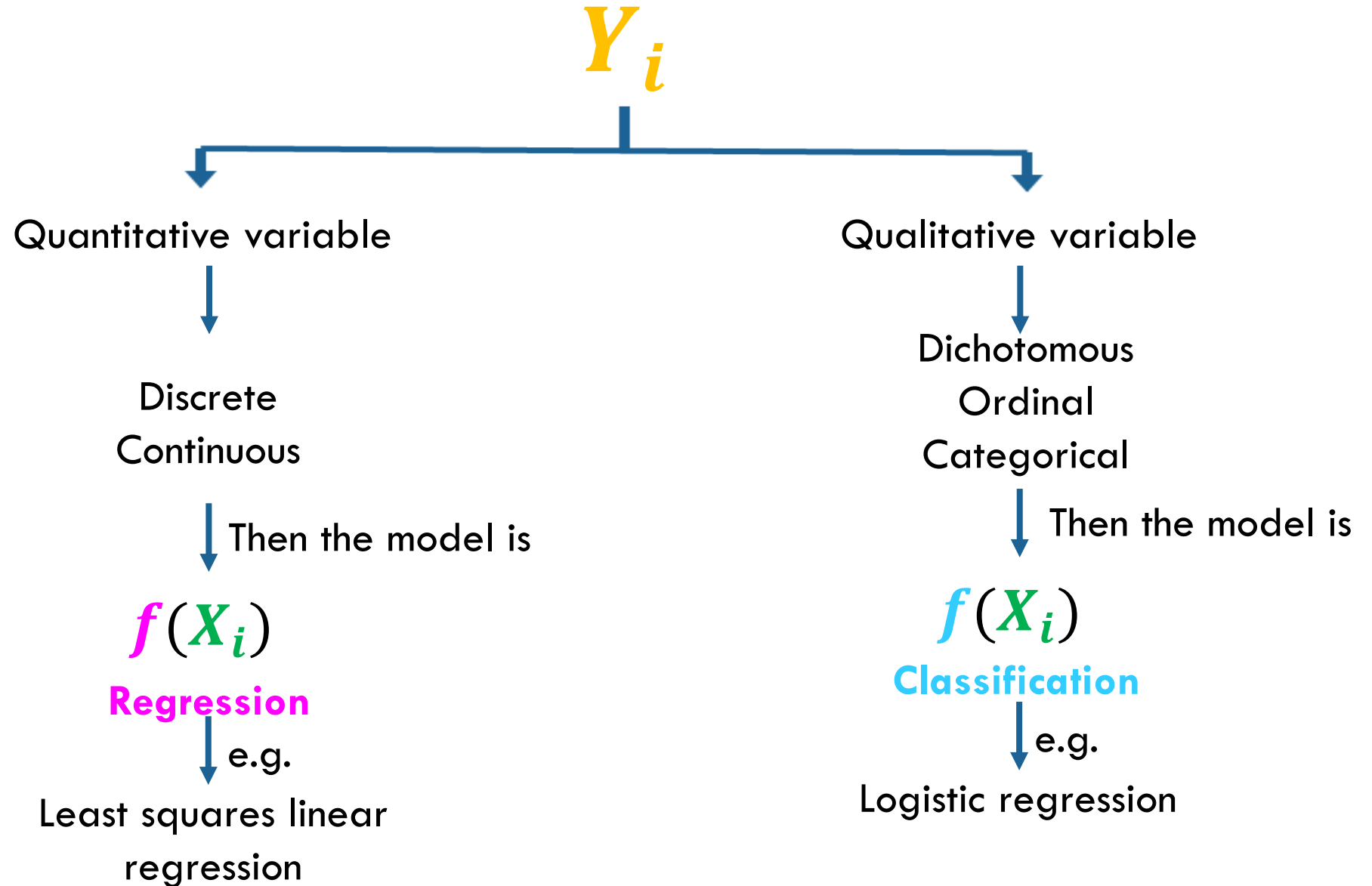


Part of exploratory data analysis.

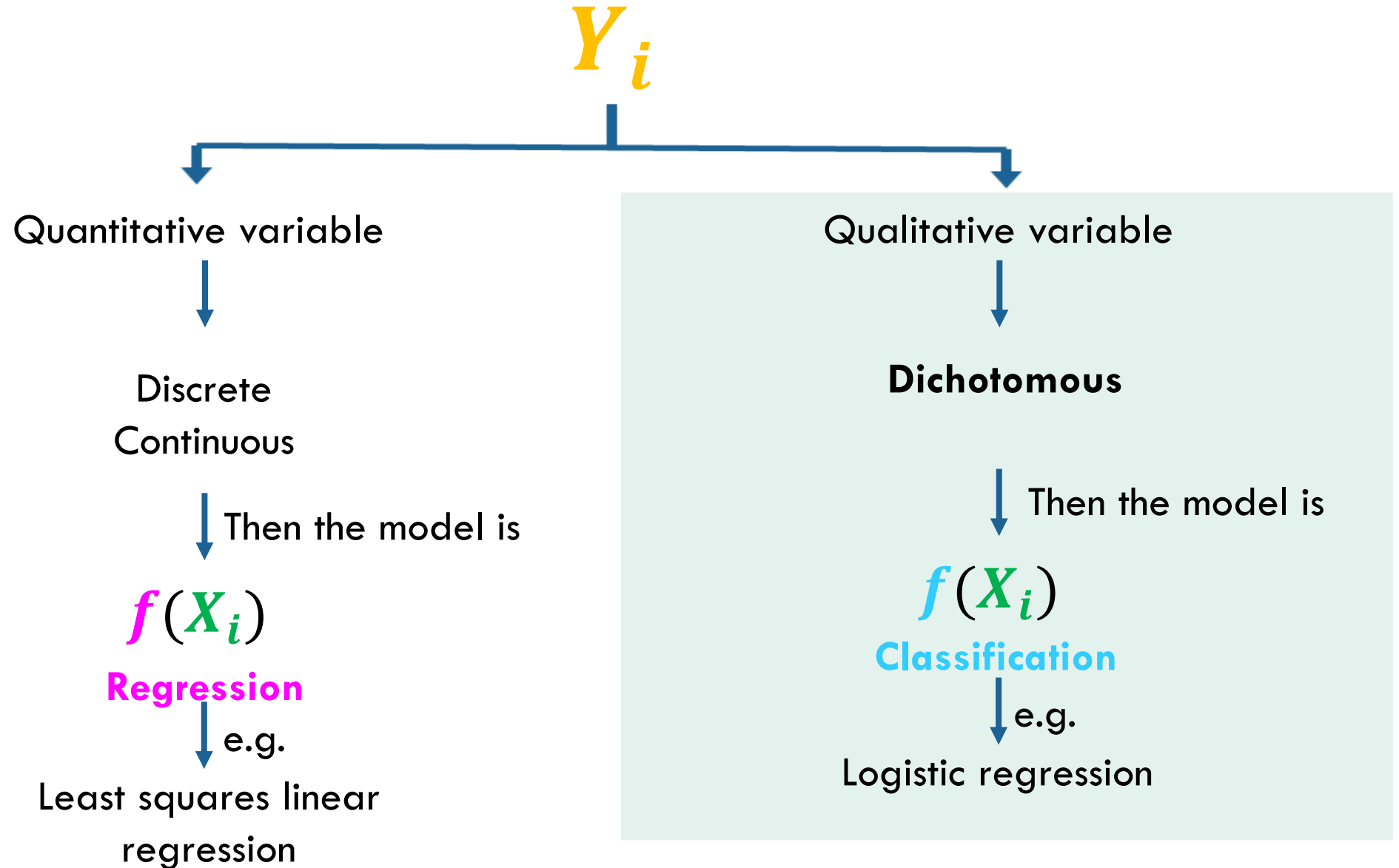
3.2 REGRESSION VS. CLASSIFICATION



REGRESSION VS. CLASSIFICATION



REGRESSION VS. CLASSIFICATION



3.3 PREDICTION VS. INFERENCE

PREDICTION VS. INFERENCE

$$Y_i \sim X1_i + X2_i + X3_i$$

↓ $f(X_i)$

$$\hat{Y}_i = \hat{\beta}_1 X1_i + \hat{\beta}_2 X2_i + \hat{\beta}_3 X3_i$$

Prediction

Predict likelihood someone gets
and allergic reaction to a
medicine.

The focus is on the predicted value:

$$\hat{Y}_i$$

We care about the **accuracy** of
the model.

Inference

Understand the **relationship between**
someone getting an allergic reaction and their
family health history.

The focus is on the coefficients:

$$\hat{\beta}_j$$

We care about the
interpretability of the model.

PREDICTION VS. INFERENCE

$$Y_i \sim X1_i + X2_i + X3_i$$

↓ $f(X_i)$

$$\hat{Y}_i = \hat{\beta}_1 X1_i + \hat{\beta}_2 X2_i + \hat{\beta}_3 X3_i$$

Prediction

Inference

Predict likelihood someone gets
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The focus is on the predicted value:

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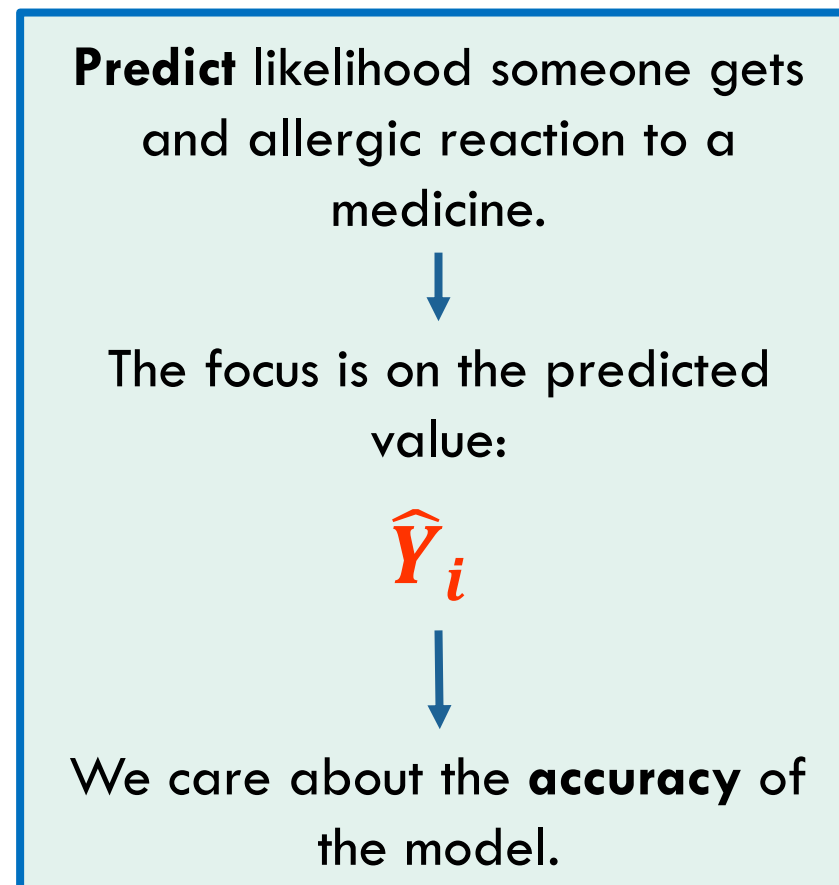
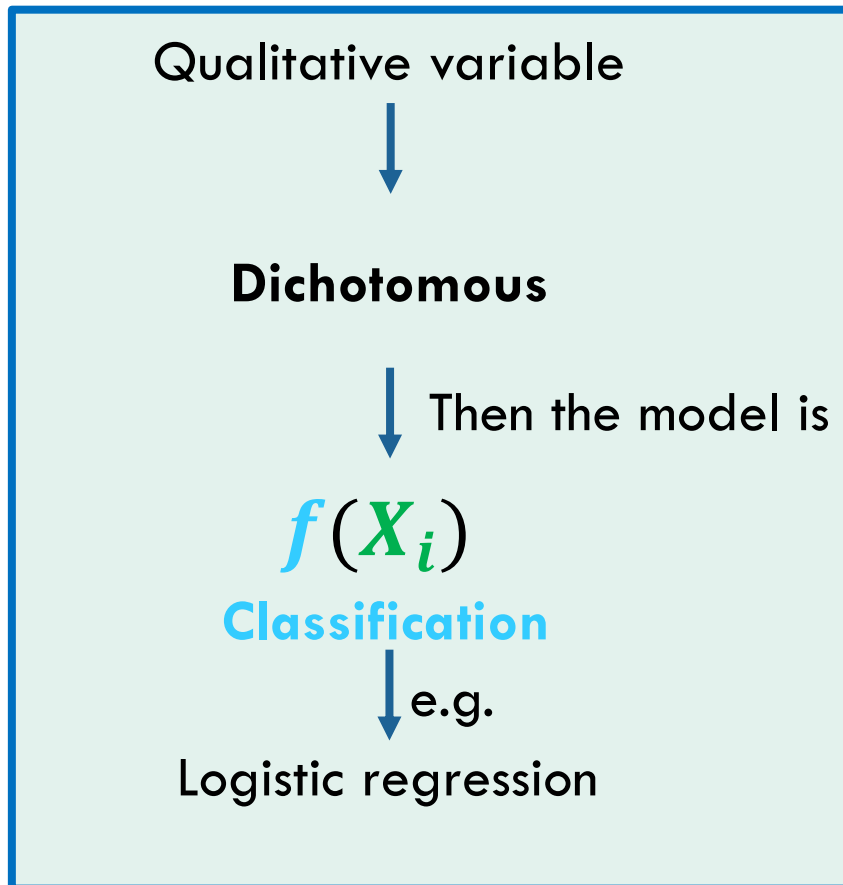
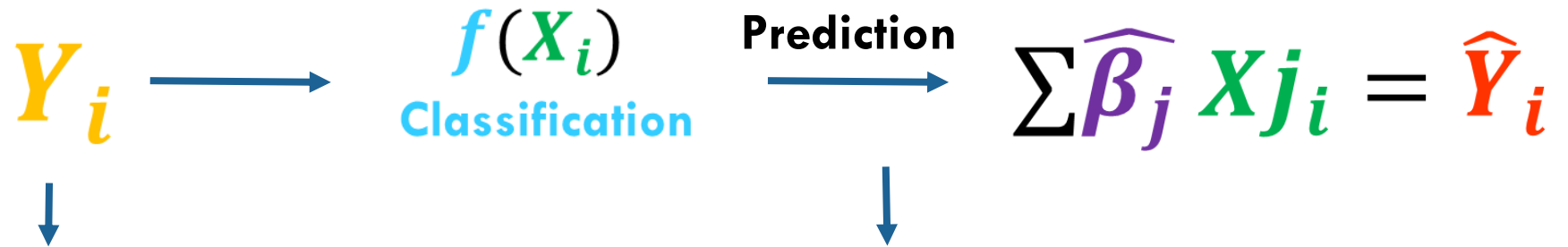
We care about the **accuracy** of
the model.

Understand the **relationship between**
someone getting an allergic reaction and their
family health history.

The focus is on the coefficients:

$$\hat{\beta}_j$$

We care about the
interpretability of the model.



3.4 THE PLAN

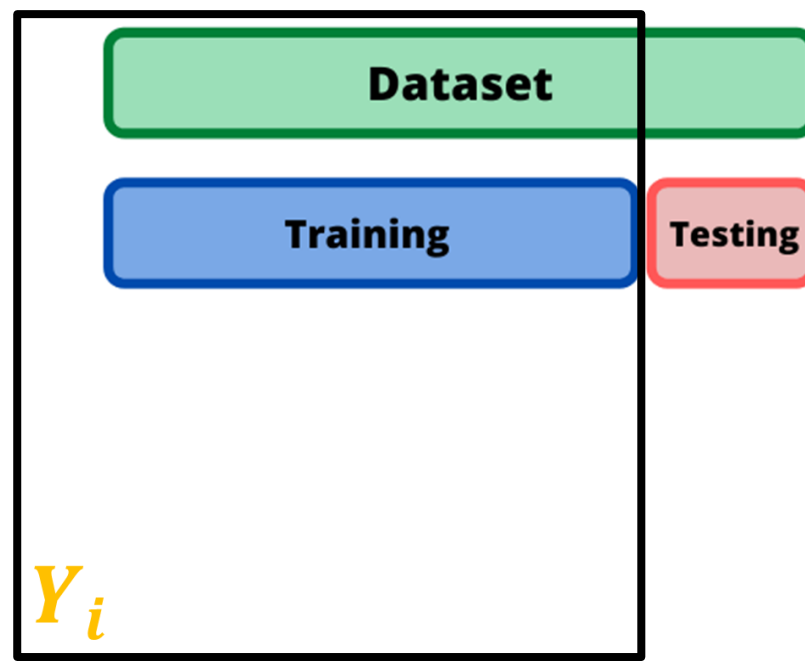
THE PLAN

Dataset

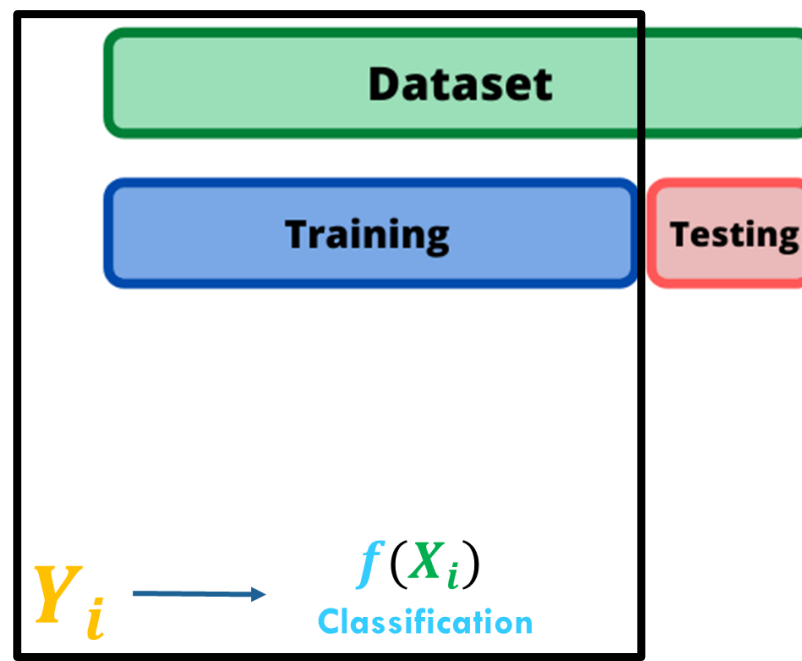
THE PLAN



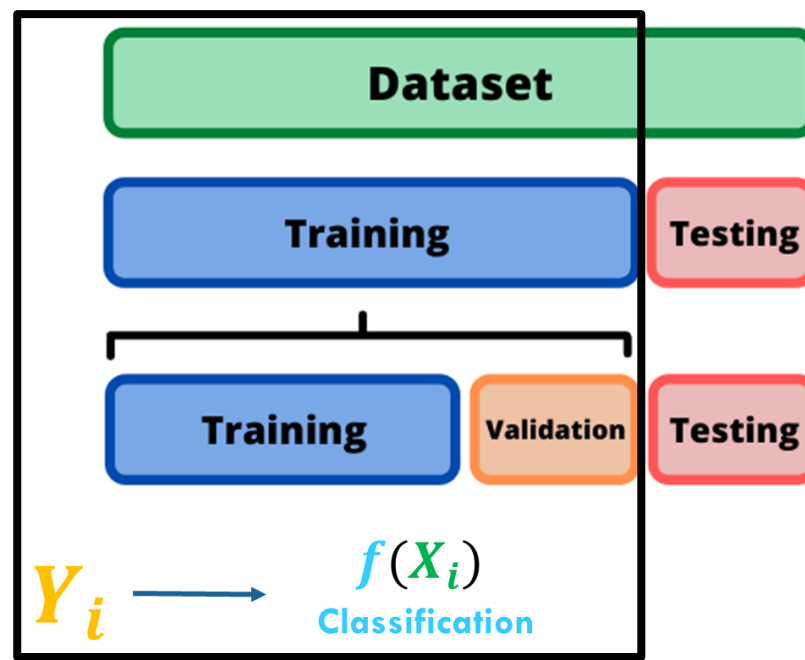
THE PLAN



THE PLAN



THE PLAN



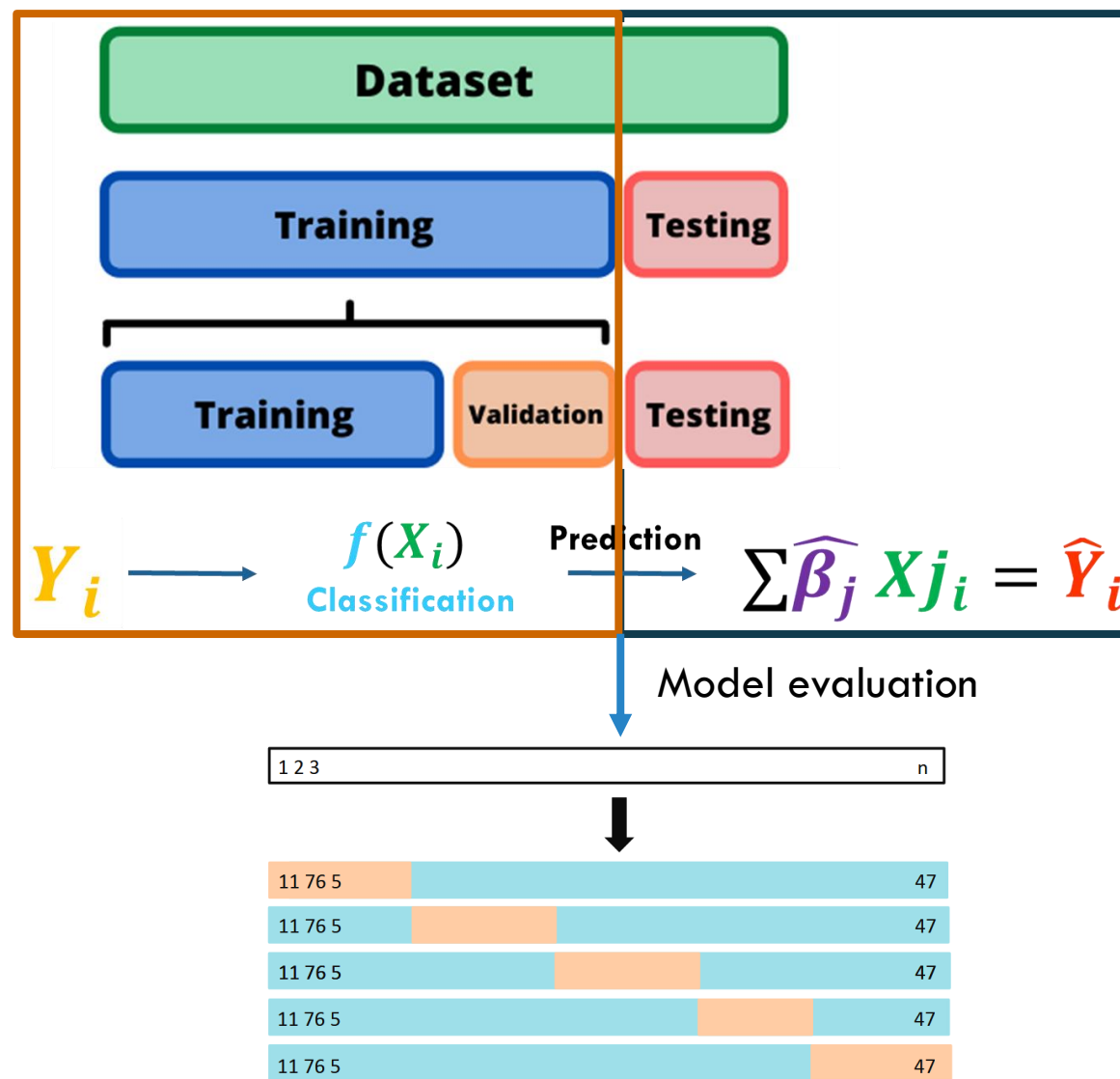
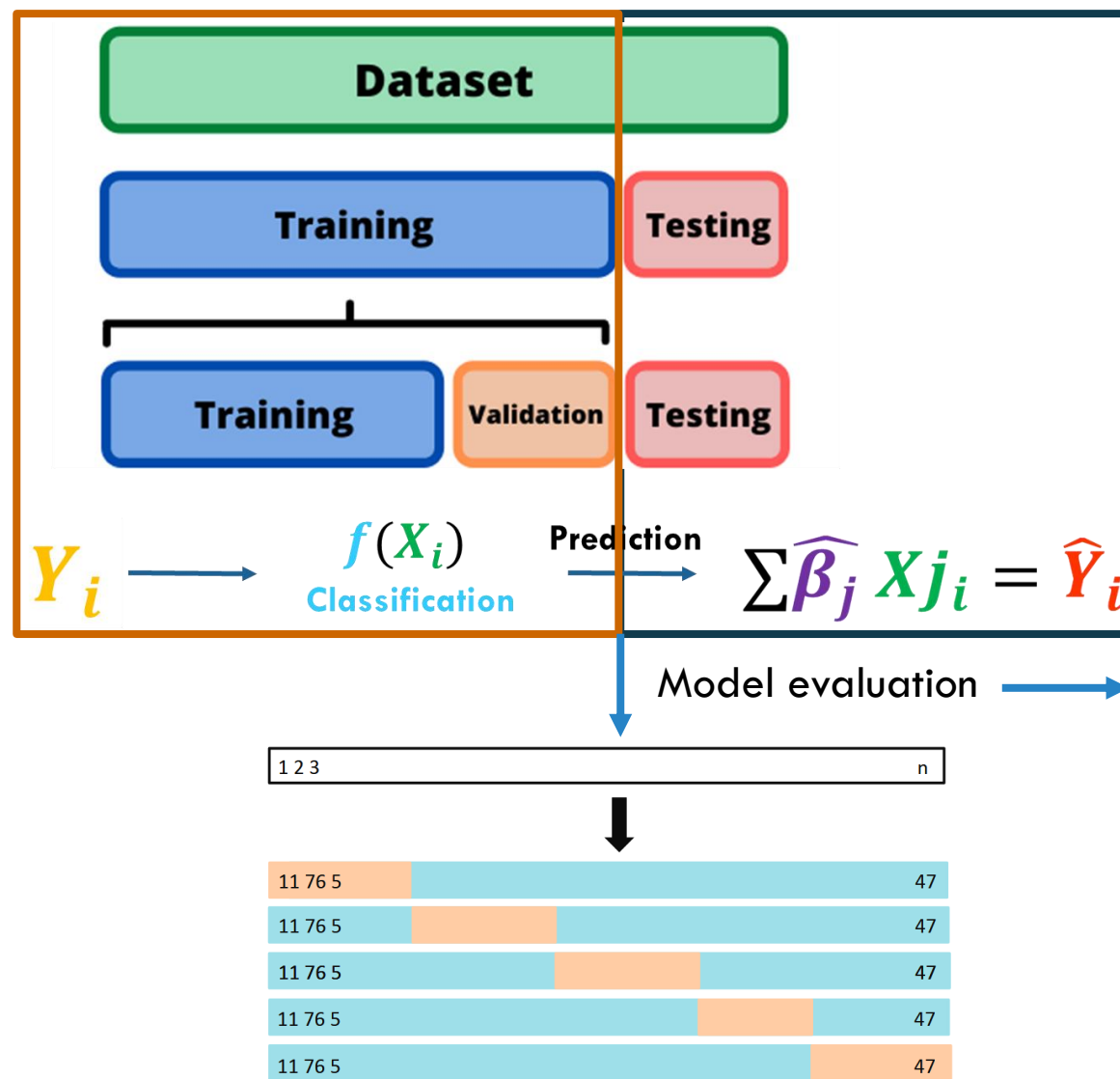


FIGURE 5.5. A schematic display of 5-fold CV. A set of n observations is randomly split into five non-overlapping groups. Each of these fifths acts as a validation set (shown in beige), and the remainder as a training set (shown in blue). The test error is estimated by averaging the five resulting MSE estimates.



Evaluation metrics for text:

- Accuracy
- Precision
- Recall
- F1-score

FIGURE 5.5. A schematic display of 5-fold CV. A set of n observations is randomly split into five non-overlapping groups. Each of these fifths acts as a validation set (shown in beige), and the remainder as a training set (shown in blue). The test error is estimated by averaging the five resulting MSE estimates.

EVALUATION METRICS FOR TEXT

Our decision of using one model or the other fully depends on the values of the following validation metrics:

- Accuracy: The overall percentage of correct predictions out of all predictions.
- Precision: Out of all the cases the model predicted as positive, how many were actually positive.
- Recall: Out of all the actual positive cases in the data, how many did the model manage to catch.
- F1-score: The harmonic mean of precision and recall. It gives a balanced view of the model's performance when you need to consider both False Positives and False Negatives.

4. TEXT CLASSIFICATION

THE DATA

```
> complaints
# A tibble: 117,214 × 18
  date_received product          sub_product issue sub_issue consumer_complaint_n...
  <date>          <chr>          <chr>      <chr> <chr>      <chr>
1 2019-09-24      Debt collection I do not k... Atte... Debt is ... "transworld systems i...
2 2019-10-25      Credit reporting, ... Credit rep... Inco... Informat... "I would like to requ...
3 2019-11-08      Debt collection I do not k... Comm... Frequent... "Over the past 2 week...
4 2019-09-08      Money transfer, vi... Domestic (... Frau... NA      "I was sold access to...
5 2019-09-24      Debt collection Medical de... Atte... Debt is ... "while checking my cr...
6 2019-11-20      credit reporting, ... Credit rep... Inco... Account ... "I would like the cre...
7 2019-09-19      Credit reporting, ... Credit rep... Inco... Personal... "MY NAME IS XXXX XXXX...
8 2019-11-22      Credit reporting, ... Credit rep... Inco... Personal... "Today XX/XX/XXXX wen...
9 2019-04-17      Credit reporting, ... Credit rep... Prob... Their in... "XXXX is reporting in...
10 2019-10-24      Credit reporting, ... Credit rep... Prob... Their in... "Please reverse the l...
# i 117,204 more rows
# i abbreviated name: 'consumer_complaint_narrative'
# i 12 more variables: company_public_response <chr>, company <chr>, state <chr>,
#   zip_code <chr>, tags <chr>, consumer_consent_provided <chr>, submitted_via <chr>,
#   date_sent_to_company <date>, company_response_to_consumer <chr>,
#   timely_response <chr>, consumer_disputed <chr>, complaint_id <dbl>
# i Use `print(n = ...)` to see more rows
```

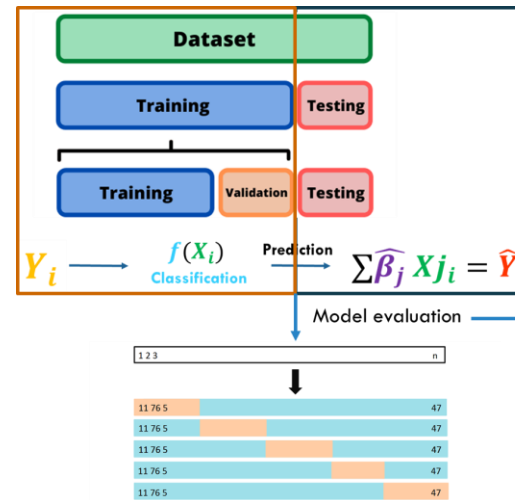
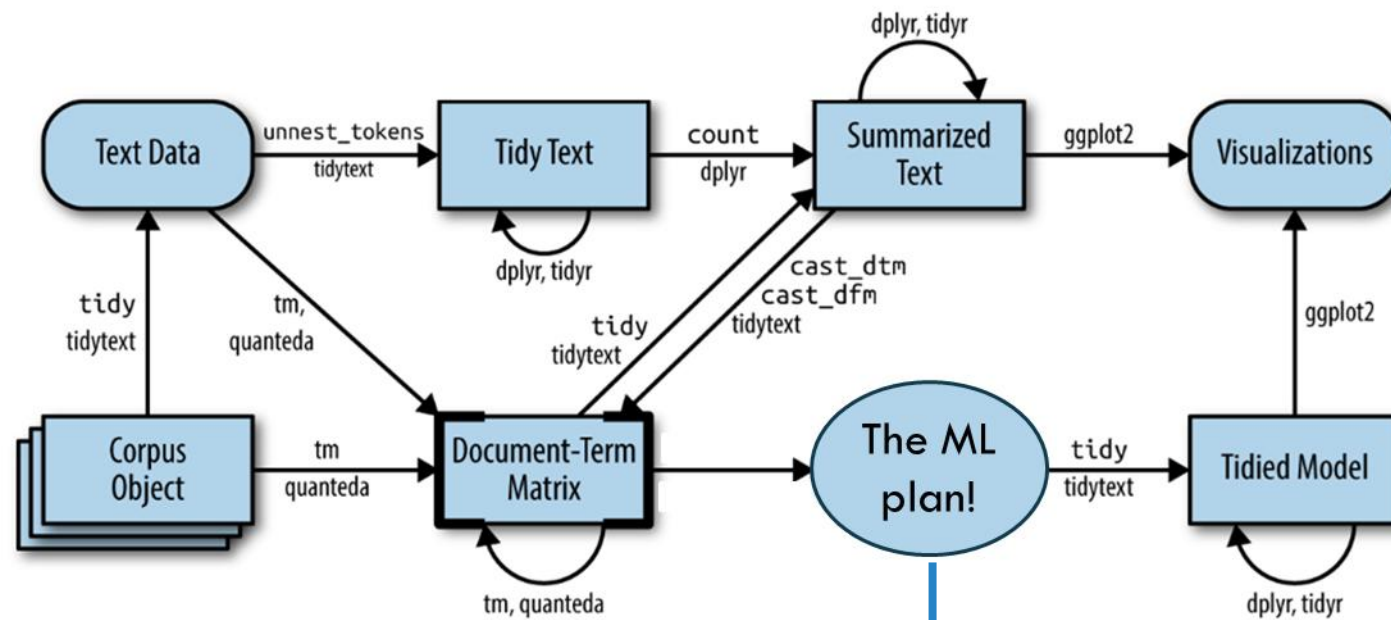
Download from here: <https://github.com/EmilHvitfeldt/smltar/blob/master/data/complaints.csv.gz>



Using R

```
complaints <-
  read_csv("https://raw.githubusercontent.com/EmilHvitfeldt/smltar/master/data/complaints.csv.gz")
```


THE PIPELINE



Evaluation metrics for text:

- Accuracy
- Precision
- Recall
- F1-score

FIGURE 5.5. A schematic display of 5-fold CV. A set of n observations is randomly split into five non-overlapping groups. Each of these fifths acts as a validation set (shown in beige), and the remainder as a training set (shown in blue). The test error is estimated by averaging the five resulting MSE estimates.



Join us!



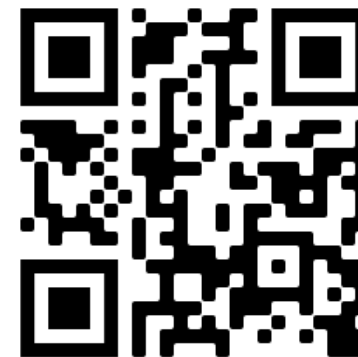
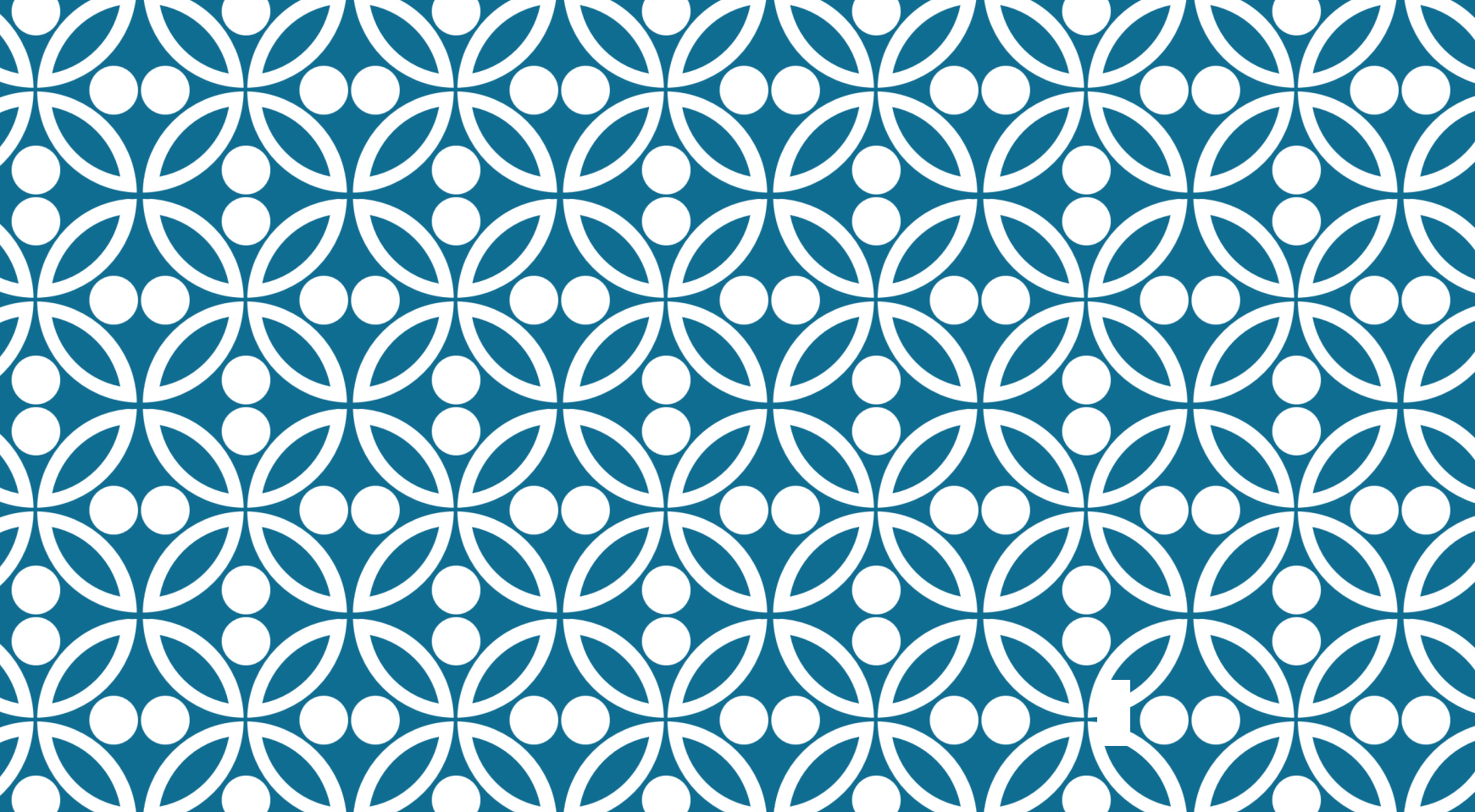
<https://forms.office.com/e/8Bgd2YsasJ>

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<https://sofiag1l.github.io/>

THANKS!

Dr. Sofia Gil-Clavel